Real-Time Perspective Correction for Selfie Video

CS 131 Final Project Milestone · Daoyuan Chi · May 2026 · Solo

1. Problem and proposal recap

Selfies and webcam video are shot at short camera-to-face distances (typically 30–60 cm), so perspective projection enlarges close-to-camera features (nose, lips, forehead) relative to farther ones (cheeks, ears, jaw). The goal of this project is a real-time pipeline that reduces that distortion *and* remains temporally stable across video frames. Prior work (Fried 2016, Shih 2019, Zhao 2019, DisCO 2024) targets single still photos; the contribution here is to push correction into a streaming, sub-real-time pipeline with explicit temporal smoothing comparison.

2. Achievements

The full Phase 0–6 pipeline is implemented and runs on a MacBook webcam at ~18–20 FPS with correction enabled. Concretely:

- **Phase 0–1**: OpenCV capture loop + MediaPipe Face Mesh (478 landmarks, iris refinement enabled). HUD shows live FPS and detection state.
- **Phase 2**: per-frame distortion metrics computed from named MediaPipe landmark indices — IPD/face_w, nose_w/face_w, nose-chin/face_w, ear-ear/face_h — logged to a per-run CSV and rendered to the HUD.
- **Phase 3 (milestone deliverable)**: *four* warp modes share the same Delaunay + `cv2.remap` infrastructure: a uniform-shrink baseline; depth-weighted radial shrink; a nose-region-only sparse warp; and the project's main contribution, a *dense depth-driven perspective re-projection* (Sec. 4). All controlled by a single `--mode` flag plus `--strength`/`--alpha`.
- **Phase 4**: alpha-mask boundary blending. The face-oval polygon is eroded by `feather/2` and Gaussian-blurred (σ=feather/2); the corrected frame is blended back onto the raw image across that band, hiding the face-outline discontinuity.
- **Phase 2×3 link (TA milestone feedback)**: `--auto-strength` derives the correction strength per frame from the measured nose-width ratio: `strength = clip(1 - target_ratio / current_ratio, 0, max)`, with target 0.22 (~85 mm portrait reference per Burgos-Artizzu 2014). Correction adapts as the user moves toward/away from the camera.
- **Phase 5**: `src/process_clip.py` records a clip and re-processes it frame-by-frame, producing `baseline.mp4` and per-frame CSV/landmark logs.
- **Phase 6**: three temporal smoothers — EMA, vectorized per-landmark Kalman, and the **1 Euro Filter** (Casiez 2012) as a third method — with a shared `update(pts, t)` interface.
- **Phase 8.5 (ahead of schedule)**: cross-subject quantitative evaluation on the Caltech Multi-Distance Portraits dataset (Burgos-Artizzu 2014), 51 subjects × 7 distances (Sec. 5).

3. Deviations from the proposal

The proposal scoped Phase 3 as a *uniform* shrink of central landmarks toward the face center by a constant factor (start with 0.92). When implemented, this visibly puffed the cheeks because the face oval was pinned while interior features compressed uniformly, leaving the cheek/jaw region disproportionately prominent. Three intermediate iterations (depth-weighted radial shrink, full-face symmetric depth re-projection, nose-region-only warp) each fixed one failure mode while introducing another; this lineage is preserved as a single `--mode` flag for ablation. The final default mode (Sec. 4) is a genuinely different formulation that the proposal did not anticipate: a *dense* warp built from MediaPipe depth, not a sparse landmark displacement. The proposal's stated metrics, smoothing methods, and evaluation plan are otherwise unchanged.

4. Method: dense depth-driven perspective re-projection

The key observation was that every sparse-landmark warp produces triangle-spanning artifacts (squashed brow, ghosting at the head outline), because the deformation lives only at the 478 landmark vertices and gets piecewise-affine-interpolated across triangles that span depth discontinuities. The new method treats MediaPipe `(x, y, z)` as a sparse 3D *depth signal* rather than a set of points to move:

1. Delaunay-triangulate over the 478 landmarks plus 8 image-corner anchors.
2. For each triangle, solve a 3×3 linear system to get an affine `z(x,y) = ax + by + c`, then rasterize `z` into a dense per-pixel depth map.
3. For every output pixel apply the pinhole perspective re-projection `scale(u,v) = α (t + 1) / (t + α)` with `t = z(u,v) / d_old`, and invert with `cv2.remap` (`source = (output − center) / scale + center`).

The control knob α = `d_new / d_old` is the virtual-camera distance ratio: α=1 is no correction, α=2 is "as if shot from twice as far," large α approaches orthographic projection. Because the depth field is smooth, the resulting warp is smooth too; there are no peripheral artifacts. Cost is comparable to the sparse warp (~44 ms per 720p frame), keeping the pipeline real-time.

5. Intermediate results

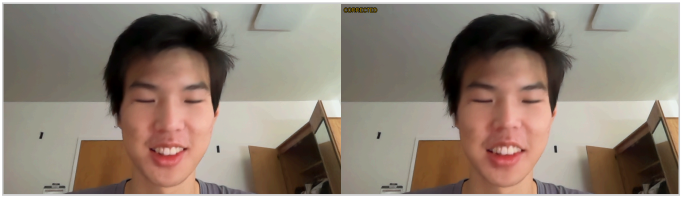

Fig. 1. Phase 3 qualitative: raw selfie (left) vs dense correction (right, α=2.0, feather=30). The face is subtly slimmer at the cheeks and the nose retreats; the head outline, hair, and background are untouched thanks to the alpha-mask boundary blend (Phase 4).

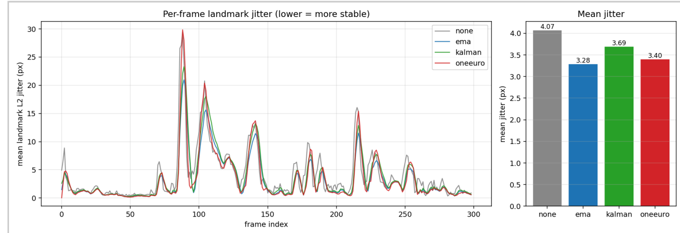

Fig. 2. Phase 6 ablation on a 10 s clip (300 frames @ 30 FPS). Mean frame-to-frame landmark L2 jitter: **raw 4.07 px** → EMA 3.28 (−19%), Kalman 3.69 (−9%), 1 Euro 3.40 (−17%). EMA is strongest on rest-state jitter but laggiest in motion; the 1 Euro Filter trades a fraction of that for better motion responsiveness, matching Casiez 2012's reported lag-jitter trade-off.

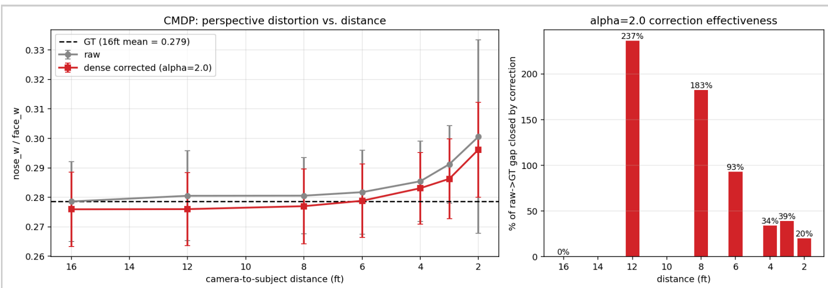

Fig. 3. Phase 8.5 (ahead of schedule): cross-subject evaluation on the Caltech Multi-Distance Portraits (Burgos-Artizzu 2014). 51 subjects × 7 distances (2–16 ft), 347 images detected and corrected. **Left**: raw nose_w/face_w inflates from 0.279 (16 ft, ground truth) to 0.301 (2 ft), confirming the metric encodes perspective distortion across subjects; the dense correction pulls the curve toward the dashed GT line at every distance. **Right**: fraction of the raw → GT gap closed by the α=2.0 correction: 20% at 2 ft, 39% at 3 ft, 93% at 6 ft. At 8–12 ft the correction overshoots, which directly motivates `--auto-strength` (Sec. 2): α should scale with measured distortion, not be a constant.

Updated timeline

Week	Status	Phases
5/15–5/22	done	0, 1, 2, 3 (4 modes), 4
5/22–5/29	done early	5, 6 (3 smoothers), **8.5**
5/29–6/3	remaining	7 (real-time opt, target 30 FPS), 8 (FPS/latency JSON), Zhao 2019 ML comparison mode, demo slides
6/3–6/6	remaining	final report (CVPR 4-page)

Risks: Zhao 2019 has CUDA-only dependencies that may not run on Mac CPU; fall-back is to skip and frame ML as future work in the report. Phase 7 is unlikely to reach 30 FPS without dropping resolution or down-sampling the per-pixel depth map; the most impactful caching is the Delaunay structure across frames once landmarks are smoothed.

6. What worked, what surprised

The auto-strength bridge between Phase 2 metrics and Phase 3 correction (per TA milestone feedback) makes the pipeline self-calibrating. CMDP confirmed the chosen metric encodes distance: across 51 subjects mean nose_w/face_w rose 7.9% from 16 ft to 2 ft. The biggest surprise was the brittleness of sparse-landmark warps — MediaPipe places landmark 10 (forehead top) far forward in z, so any radial-shrink moved it 50–65 px and distorted the head. The dense reformulation made it go away.

7. Repository

Code, plots, and per-frame CSV/landmark logs at `github.com/Danni281/cs-131-final-project`. Headline figures: `results/phase6_jitter.png`, `results/cmdp_eval.png`. Default mode: `python src/main.py --correct-on-start --alpha 2.0`; every variant exposed via CLI flags for ablation. Dev log: `CLAUDE.md`; phase tracker: `README.md`.